

TIAN HONG RAYMOND WU

[Z5360249]

COMP9417 26T2

HOMEWORK 2

Created 3rd July, 2026

[Note: Each Question begins on a new page.]

Appendix at end of report

Problems will reference any of the derivations by stating the number label e.g. "according to [1]"

Question 1 (a)

$$\hat{T}_1 = a\bar{X} + (1-a) \cdot c_* \cdot S$$

$$\text{Bias}[\hat{T}_1] = E[\hat{T}_1] - \gamma$$

$$= E[a\bar{X} + (1-a) \cdot c_* \cdot S] - \gamma$$

$$\begin{aligned} E[aX] &= a E[X] \\ E[X+Y] &= E[X] + E[Y] \end{aligned}$$

$$= a E[\bar{X}] + (1-a) E[c_* \cdot S] - \gamma$$

$$= a\gamma + (1-a)\gamma - \gamma$$

see Appendix [1]
near end of report

$$= a\gamma + \gamma - a\gamma - \gamma$$

$$= 0 //$$

Hence, $\text{Bias}[\hat{T}_1] = 0$, so $\hat{T}_1$ is an unbiased estimator of γ . //

Question 1 (b) Find parameter 'a' that gives least mean squared error (MSE) for $\hat{T}_2$.

Mean Squared Error $MSE(\hat{T}_2) = E[(\hat{T}_2 - \delta)^2]$ i.e. long-term average of the error or difference between the estimated mean and the true mean.
 $= Bias[\hat{T}_2]^2 + Var[\hat{T}_2]$

$$MSE(\hat{T}_2) = Bias[\hat{T}_2]^2 + Var[\hat{T}_2]$$

$$= Var[\hat{T}_2]$$

$$= Var[a\bar{X} + (1-a)c_* \cdot S]$$

$$= a^2 Var[\bar{X}] + (1-a)^2 \cdot Var[c_* \cdot S]$$

$$= a^2 \frac{\sigma^2}{n} + (1-a)^2 \cdot \sigma^2 \cdot (c_*^2 - 1)$$

see [3] & [8]

To find $\underset{a}{\operatorname{argmin}} MSE(\hat{T}_2)$ i.e. value of 'a' that produces the minimum $MSE(\hat{T}_2)$.

$$\frac{dMSE(\hat{T}_2)}{da} = 0$$

Find the minimum (or maximum but we will prove via second derivative test) of its derivative which signifies min (or max) error.

Chain Rule

$$g(a) = 1-a \quad h(g(a)) = g(a)^2 \\ g'(a) = -1 \quad h'(g(a)) = 2g(a)$$

$$\Rightarrow 2a \frac{\sigma^2}{n} - 2(1-a) \sigma^2 (c_*^2 - 1) = 0$$

$$\Rightarrow a \cdot \frac{1}{n} + a(c_*^2 - 1) = (c_*^2 - 1)$$

$$\Rightarrow a \left[\frac{1 + n \cdot (c_*^2 - 1)}{n} \right] = c_*^2 - 1$$

$$\Rightarrow a = \frac{nc_*^2 - n}{nc_*^2 - n + 1} //$$

Check that 'a' produces minimum $MSE(\hat{T}_2^*)$ (second derivative test)

$$\frac{d^2 MSE(\hat{T}_2)}{da^2} = 2 \frac{\sigma^2}{n} + 2 \sigma^2 (c_*^2 - 1)$$

$$= 2 \sigma^2 \left(\frac{1}{n} + c_*^2 - 1 \right) > 0$$

to satisfy double derivative test for minimum.

The chosen value 'a' is only a minimum if $\left(\frac{1}{n} + c_*^2 \right) > 1$
 $\Rightarrow c_*^2 > 1 - \frac{1}{n}$

Hence, we have found $\hat{T}_2^*$ where 'a' is chosen to make $\hat{T}_2$ the best possible estimator via MSE method.

$$\hat{T}_2^* = \frac{nc_*^2 - n}{nc_*^2 - n + 1} \bar{X} + \left(1 - \frac{nc_*^2 - n}{nc_*^2 - n + 1} \right) \cdot c_* \cdot S \\ = \frac{nc_*^2 - n}{nc_*^2 - n + 1} \bar{X} + \left(\frac{1}{nc_*^2 - n + 1} \right) \cdot c_* \cdot S //$$

Question 1 (c) Find parameters a_1 and a_2 that minimises the MSE of estimator $\hat{T}_2$.

$$\hat{T}_2 = a_1 \bar{X} + a_2 \cdot C_x \cdot S$$

Mean Squared Error $MSE(\hat{T}_2) = E[(\hat{T}_2 - \gamma)^2]$ i.e. long-term average of the error or difference between the estimated mean and the true mean.
 $= \text{Bias}[\hat{T}_2]^2 + \text{Var}[\hat{T}_2]$
 This method works as well.

$$\begin{aligned} MSE(\hat{T}_2) &= E[(\hat{T}_2 - \gamma)^2] \\ &= E[\hat{T}_2^2 - 2\gamma\hat{T}_2 + \gamma^2] \\ &= \underline{E[\hat{T}_2^2]} - 2\gamma \underline{E[\hat{T}_2]} + \gamma^2 \quad \text{--- (A)} \end{aligned}$$

$$\begin{aligned} E[\hat{T}_2^2] &= E[(a_1 \bar{X} + a_2 \cdot C_x \cdot S)^2] \\ &= E[a_1^2 \bar{X}^2 + 2 \cdot a_1 a_2 \cdot \bar{X} \cdot C_x \cdot S + a_2^2 \cdot C_x^2 \cdot S^2] \\ &= a_1^2 E[\bar{X}^2] + 2a_1 a_2 E[\bar{X} \cdot C_x \cdot S] + a_2^2 E[C_x^2 \cdot S^2] \\ &= a_1^2 \cdot \gamma^2 \left(\frac{1}{n} + 1\right) + 2a_1 a_2 \gamma^2 + a_2^2 \cdot \gamma^2 \cdot C_x^2 \quad \text{--- (B)} \end{aligned}$$

see [1], [4], [7]

$$\begin{aligned} E[\hat{T}_2] &= E[a_1 \bar{X} + a_2 \cdot C_x \cdot S] \\ &= a_1 \gamma + a_2 \gamma \quad \text{--- (C)} \end{aligned}$$

see [1]

sub (B) & (C) into (A):

$$MSE(\hat{T}_2) = a_1^2 \cdot \gamma^2 \left(\frac{1}{n} + 1\right) + 2a_1 a_2 \gamma^2 + a_2^2 \gamma^2 \cdot C_x^2 - 2a_1 \gamma^2 - 2a_2 \gamma^2 + \gamma^2$$

Find $\arg\min_{a_1} MSE(\hat{T}_2)$

$$\frac{dMSE(\hat{T}_2)}{da_1} = 0$$

$$2a_1 \gamma^2 \left(\frac{1}{n} + 1\right) + 2a_2 \gamma^2 - 2\gamma^2 = 0$$

$$\Rightarrow a_1 \left(\frac{1}{n} + 1\right) = 1 - a_2$$

$$\Rightarrow a_1 + n \cdot a_1 = n - n \cdot a_2$$

$$\Rightarrow a_1 = \frac{n - n \cdot a_2}{n + 1} \quad \text{--- (D)}$$

Find $\arg\min_{a_2} \text{MSE}(\hat{T}_2)$

$$\frac{\partial \text{MSE}(\hat{T}_2)}{\partial a_2} = 0$$

$$2a_2 \gamma^2 + 2a_2 \gamma^2 C_x^2 - 2\gamma^2 = 0$$

$$\Rightarrow a_2 C_x^2 = 1 - a_1$$

$$\Rightarrow a_2 = \frac{1-a_1}{C_x^2} \quad \text{--- (E)}$$

substitute (E) into (D):

$$n a_1 + a_1 = n - n \cdot \frac{1-a_1}{C_x^2}$$

$$C_x^2(n \cdot a_1 + a_1) = n \cdot C_x^2 - n + n \cdot a_1$$

$$\Rightarrow n \cdot C_x^2 \cdot a_1 + C_x^2 a_1 - n \cdot a_1 = n \cdot C_x^2 - n$$

$$\Rightarrow a_1 (n \cdot C_x^2 + C_x^2 - n) = n \cdot C_x^2 - n$$

$$\Rightarrow a_1 = \frac{n \cdot C_x^2 - n}{n \cdot C_x^2 + C_x^2 - n} //$$

substitute (D) into (E)

$$C_x^2 a_2 = 1 - \frac{n - n \cdot a_2}{n+1}$$

$$\Rightarrow C_x^2 a_2 = \frac{n+1 - n + n \cdot a_2}{n+1}$$

$$\Rightarrow C_x^2 a_2 (n+1) = 1 + n \cdot a_2$$

$$\Rightarrow C_x^2 \cdot a_2 \cdot n + C_x^2 \cdot a_2 - n \cdot a_2 = 1$$

$$\Rightarrow a_2 (C_x^2 \cdot n + C_x^2 - n) = 1$$

$$\Rightarrow a_2 = \frac{1}{C_x^2 \cdot n + C_x^2 - n} //$$

Check that ' a_1 ' produces minimum $MSE(\hat{T}_2^*)$ (second derivative test)

$$\frac{d^2 MSE(\hat{T}_2)}{da_1^2} = 2\gamma^2 + 2\frac{\gamma^2}{n} > 0 \quad \text{always since } \gamma > 0$$

The chosen a_1 is a minimum for $MSE(\hat{T}_2)$.

Check that ' a_2 ' produces minimum $MSE(\hat{T}_2^*)$

$$\begin{aligned} \frac{d^2 MSE(\hat{T}_2)}{da_2^2} &= 2\gamma^2 + 2\gamma^2(C_*^2 - 1) \\ &= 2\gamma^2 C_*^2 > 0 \quad \text{always since } C_*^2 > 0, \gamma^2 > 0 \end{aligned}$$

The chosen a_2 is a minimum for $MSE(\hat{T}_2)$.

Hence, we have found a_1 & a_2 that minimizes the $MSE(\hat{T}_2)$ and gives the best possible estimator for $\hat{T}_2$.

$$\hat{T}_2^* = \frac{nC_*^2 - n}{nC_*^2 - n + C_*^2} \cdot \bar{X} + \frac{1}{nC_*^2 - n + C_*^2} \cdot C_* \cdot S$$

Question 1 (d) Show that $MSE(\hat{T}_2^*) \leq MSE(\hat{T}_1^*)$

$$\hat{T}_1^* = \frac{nC_*^2 - n}{nC_*^2 - n + 1} \bar{X} + \left(\frac{1}{nC_*^2 - n + 1} \right) C_* \cdot S$$

$$= A \bar{X} + B C_* \cdot S$$

$$\hat{T}_2^* = \frac{nC_*^2 - n}{nC_*^2 - n + C_*^2} \cdot \bar{X} + \frac{1}{nC_*^2 - n + C_*^2} \cdot C_* \cdot S$$

$$= M \bar{X} + N C_* \cdot S$$

$$MSE(\hat{T}_1^*) = E[(T_1^* - \gamma)^2]$$

$$= E[T_1^{*2} - 2\gamma T_1^* + \gamma^2]$$

$$= \underline{E[T_1^{*2}]} - 2\gamma \underline{E[T_1^*]} + \gamma^2 \quad \text{--- (1)}$$

$$E[T_1^{*2}] = E[(A\bar{X} + B C_* \cdot S)^2]$$

$$= E[A^2 \bar{X}^2 + 2AB \bar{X} C_* \cdot S + B^2 C_*^2 S^2]$$

$$= A^2 \gamma^2 \left(1 + \frac{1}{n}\right) + 2AB \gamma^2 + B^2 C_*^4 \gamma^2$$

$$E[T_1^*] = A\gamma + B\gamma$$

$$\therefore MSE(\hat{T}_1^*) = A^2 \gamma^2 \left(1 + \frac{1}{n}\right) + 2AB \gamma^2 + B^2 C_*^4 \gamma^2 - 2\gamma^2 A - 2\gamma^2 B + \gamma^2$$

$$= A^2 \gamma^2 + A^2 \gamma^2 \frac{1}{n} + 2AB \gamma^2 + B^2 C_*^4 \gamma^2 - 2\gamma^2 A - 2\gamma^2 B + \gamma^2$$

Similarly,

$$MSE(\hat{T}_2^*) = M^2 \gamma^2 \left(1 + \frac{1}{n}\right) + 2M \cdot N \gamma^2 + N^2 C_*^4 \gamma^2 - 2\gamma^2 M - 2\gamma^2 N + \gamma^2$$

We can see that $-2\gamma^2 M - 2\gamma^2 N < -2\gamma^2 A - 2\gamma^2 B$ since we expect $C_* \leq 1$ (denominator of M & N is bigger than that of A & B i.e. $M, N > A, B$).

Question 1 (e)

$$\hat{V}_+ = \max \{0, \hat{T}_2^*\} \quad \text{Let } \gamma \text{ be the true paramtr.}$$

$$\text{when } \hat{T}_2^* \geq 0, \quad V_+ = \max \{0, \hat{T}_2^*\} = \hat{T}_2^* \quad \text{Squared Error: } (\hat{V}_+ - \gamma)^2 = (\hat{T}_2^* - \gamma)^2$$

$$\text{when } \hat{T}_2^* < 0, \quad V_+ = \max \{0, \hat{T}_2^*\} = 0 \quad \text{Squared Error: } (\hat{V}_+ - \gamma)^2 = (0 - \gamma)^2 = \gamma^2$$

$$\text{Consider when } \gamma > 0, \quad \hat{T}_2^* \leq 0, \quad \hat{T}_2^* \leq 0 < \gamma$$

$$\Rightarrow \hat{T}_2^* - \gamma \leq -\gamma < \gamma - \gamma$$

$$\Rightarrow \hat{T}_2^* - \gamma \leq -\gamma < 0$$

$$\Rightarrow (\hat{T}_2^* - \gamma)^2 \geq \gamma^2$$

$$\Rightarrow \gamma^2 = (V_+ - \gamma)^2 \Rightarrow (\hat{T}_2^* - \gamma)^2 \geq (V_+ - \gamma)^2$$

$$\Rightarrow E[(\hat{T}_2^* - \gamma)^2] \geq E[(V_+ - \gamma)^2]$$

$$\Rightarrow \text{MSE}(\hat{T}_2^*) \geq \text{MSE}(V_+) //$$

flip sign since
 $\hat{T}_2^* \leq 0$. e.g.
 $T_2^* = -2, \gamma = -3,$

$$-2 - (-3) = -1$$

$$(-1)^2 = 1 \geq (-3)^2 = 9$$

Question 1 (f)

$$\hat{T}_3 = a_1 \bar{X} + a_2 c_{*} S + a_3$$

$$\hat{T}_3 \text{ is unbiased i.e. } \text{Bias}[\hat{T}_3] = E[\hat{T}_3] - \gamma = 0 \\ \Rightarrow E[\hat{T}_3] = \gamma$$

$$\text{Cannot make assumptions about } a_1, a_2, a_3 \text{ i.e. } \text{Var}[a_1 + a_2 + a_3] \neq \text{Var}[a_1] + \text{Var}[a_2] + \text{Var}[a_3]$$

$$\text{MSE}(\hat{T}_3) = E[(\hat{T}_3 - \gamma)^2]$$

$$= E[\hat{T}_3^2 - 2\gamma\hat{T}_3 + \gamma^2]$$

$$= E[\hat{T}_3^2] - 2\gamma E[\hat{T}_3] + \gamma^2 \quad E[\hat{T}_3] = \gamma$$

$$= E[(a_1 \bar{X} + a_2 c_{*} S + a_3)^2] - 2\gamma^2 + \gamma^2$$

$$= E[a_1^2 \bar{X}^2 + a_2^2 c_{*}^2 S^2 + a_3^2 + 2a_1 a_2 \bar{X} c_{*} S + 2a_1 a_3 \bar{X} + 2a_2 a_3 c_{*} S] - \gamma^2$$

$$= a_1^2 \gamma^2 \left(\frac{1}{n} + 1\right) + a_2^2 c_{*}^2 \gamma^2 + a_3^2 + 2a_1 a_2 \gamma^2 + 2a_1 a_3 \gamma + 2a_2 a_3 \gamma - \gamma^2$$

$$\frac{\partial \text{MSE}(\hat{T}_3)}{\partial a_1} = 2a_1 \gamma^2 \left(\frac{1}{n} + 1\right) + 2a_2 \gamma^2 + 2a_3 \gamma = 0$$

$$\frac{\partial \text{MSE}(\hat{T}_3)}{\partial a_2} = 2a_1 \gamma^2 + 2a_2 c_{*}^2 \gamma^2 + 2a_3 \gamma = 0$$

$$\frac{\partial \text{MSE}(\hat{T}_3)}{\partial a_3} = 2a_1 \gamma + 2a_2 \gamma + 2a_3 = 0$$

$$\text{Solve for } a_1, a_2, a_3 \quad \begin{bmatrix} a_1 & a_2 & a_3 \\ 2\gamma^2(\frac{1}{n}+1) & 2\gamma^2 & 2\gamma \\ 2\gamma^2 & 2c_{*}^2\gamma^2 & 2\gamma \\ 2\gamma & 2\gamma & 2 \end{bmatrix} \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$$

INCOMPLETE

Question 2 (a)

$$\ell(y, \hat{y}) = \frac{1}{2} (y - \hat{y})^2$$

For round 't' of algorithm, show that $r_{t,i} = y_i - f_{t-1}(x_i)$

$$r_{t,i} = -\frac{\partial}{\partial f(x_i)} \sum_{j=1}^n \ell[y_j, f(x_j)] \Big|_{f(x_j) = f_{t-1}(x_j), j=1, \dots, n}$$

$$= -\frac{\partial}{\partial f(x_i)} \sum_{j=1}^n \frac{1}{2} [y_j - f(x_j)]^2 \Big|_{f(x_j) = f_{t-1}(x_j), j=1, \dots, n}$$

$$= -\frac{\partial}{\partial f(x_i)} \frac{1}{2} [(y_1 - f(x_1))^2 + \dots + (y_i - \underline{\hat{f}(x_i)})^2 + \dots + (y_n - f(x_n))^2] \Big|_{\substack{f(x_j) = f_{t-1}(x_j) \\ f(x_i) = \hat{f}(x_i)}}$$

$$= -\frac{1}{2} [-2(y_i - \hat{f}(x_i))] \Big|_{f(x_j) = f_{t-1}(x_j)}$$

$$= y_i - \hat{f}(x_i) \Big|_{f(x_j) = f_{t-1}(x_j)}$$

$$= y_i - f_{t-1}(x_i) //$$

$$(GC3) \quad h_t = \argmin_{f \in \mathcal{F}} \sum_{i=1}^n \ell(r_{t,i}, f(x_i))$$

$$= \argmin_{f \in \mathcal{F}} \sum_{i=1}^n \frac{1}{2} [y_i - f_{t-1}(x_i) - f(x_i)]^2 //$$

$$= \argmin_{f \in \mathcal{F}} \sum_{i=1}^n \frac{1}{2} [y_i^2 - 2y_i f_{t-1}(x_i) - 2y_i f(x_i) + 2f_{t-1}(x_i) f(x_i) + f(x_i)^2 + f_{t-1}(x_i)^2]$$

Question 2 (b)

$$\alpha_+ = \arg \min_{\alpha} \sum_{i=1}^n \ell[y_i, f_{+-}(x_i) + \alpha h_+(x_i)]$$

$$= \arg \min_{\alpha} \sum_{i=1}^n \frac{1}{2} \left[y_i - \underbrace{f_{+-}(x_i)}_{r_{+,i}} - \alpha h_+(x_i) \right]^2$$

$$= \arg \min_{\alpha} \sum_{i=1}^n \frac{1}{2} [r_{+,i} - \alpha h_+(x_i)]^2$$

$$= \arg \min_{\alpha} \sum_{i=1}^n \frac{1}{2} (r_{+,i}^2 - 2 r_{+,i} \alpha h_+(x_i) + \alpha^2 h_+(x_i)^2)$$

Minimize:

$$\begin{aligned} \frac{d\alpha_+}{d\alpha} &= \frac{1}{2} \sum_{i=1}^n \left[-2 r_{+,i} h_+(x_i) \frac{d(\alpha)}{d\alpha} \right] + \frac{1}{2} \sum_{i=1}^n \left[h_+(x_i)^2 \cdot \frac{d(\alpha^2)}{d\alpha} \right] \\ &= - \sum_{i=1}^n r_{+,i} \cdot h_+(x_i) + \sum_{i=1}^n h_+(x_i)^2 \cdot \alpha \end{aligned}$$

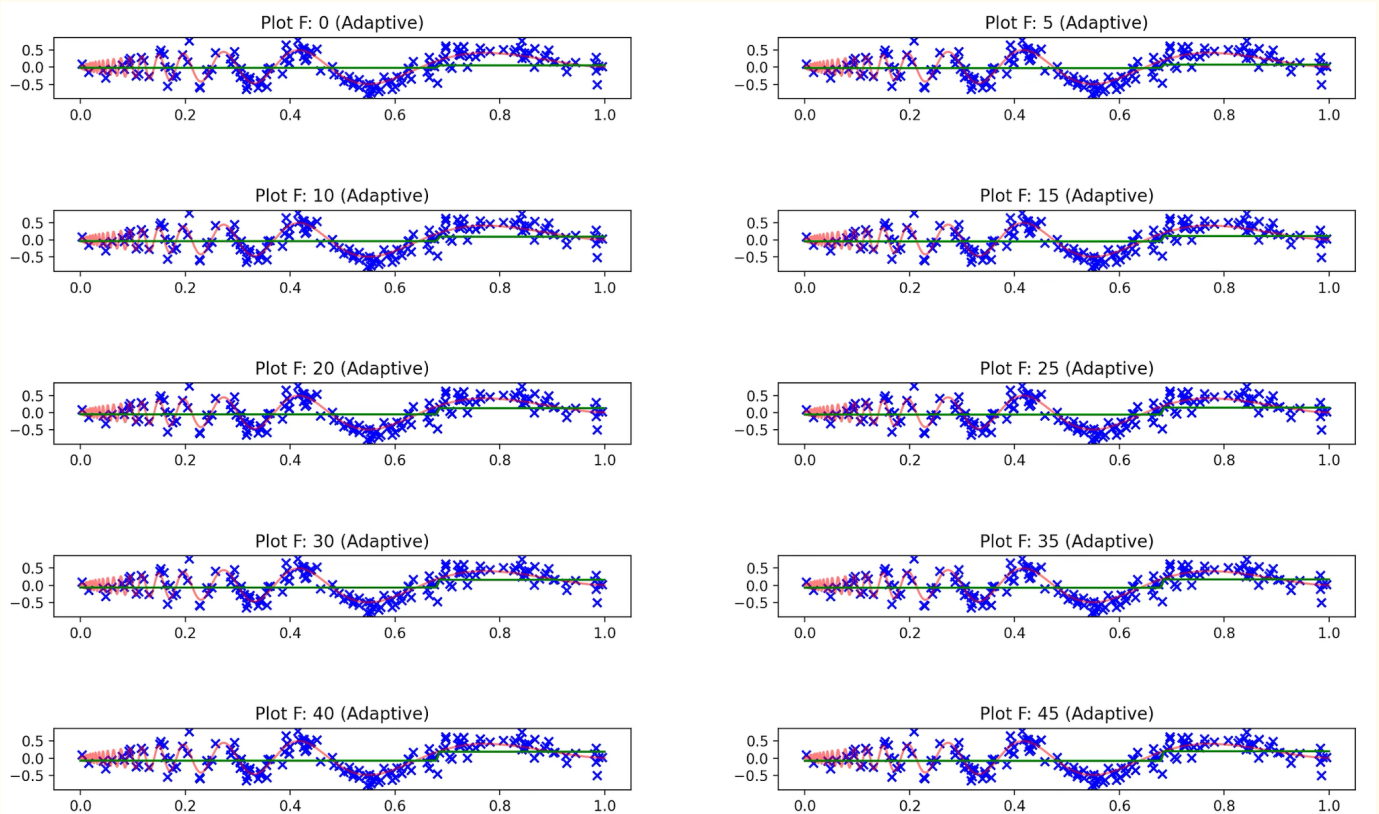
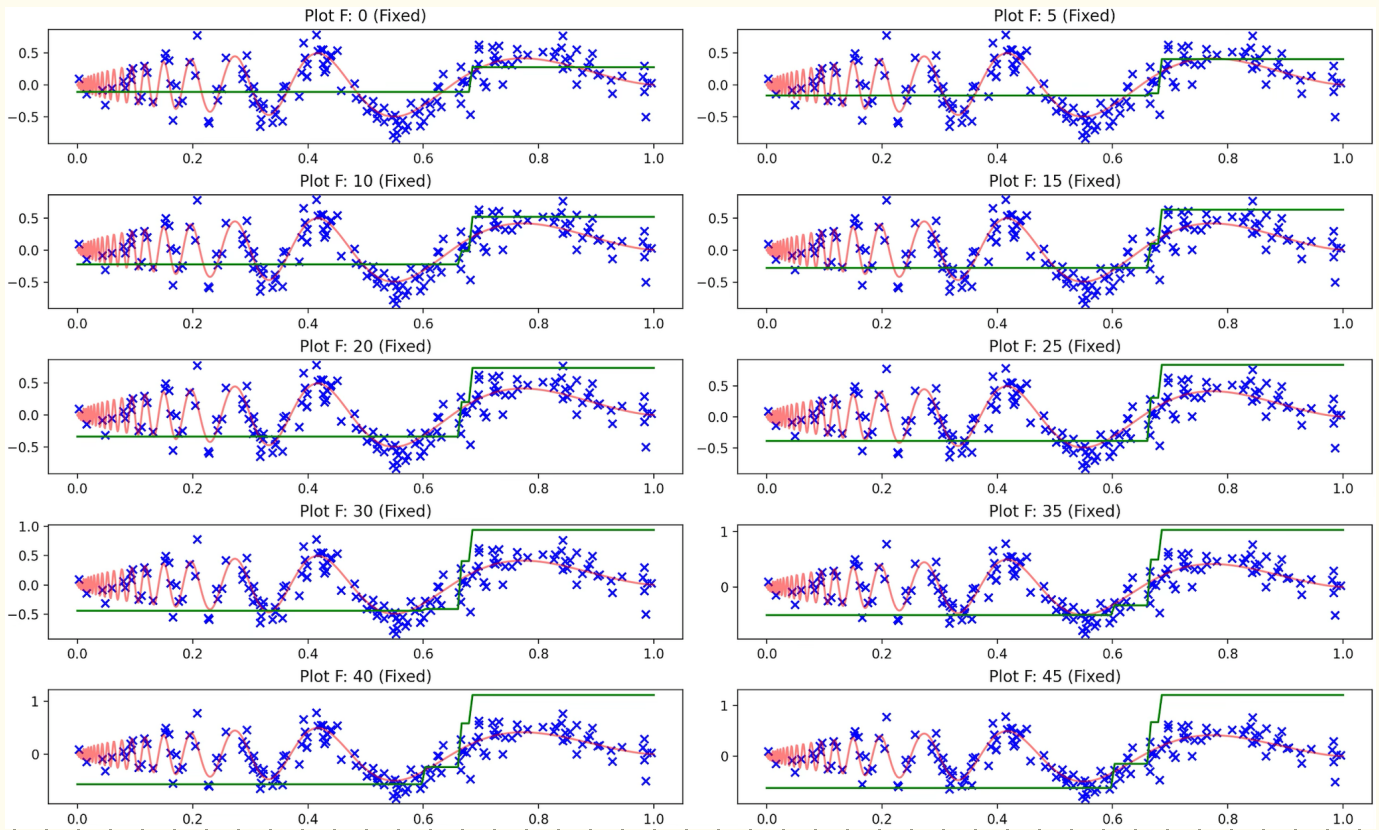
$$\frac{d\alpha_+}{d\alpha} = 0$$

$$\Rightarrow \alpha = \frac{\sum_{i=1}^n r_{+,i} h_+(x_i)}{\sum_{i=1}^n h_+(x_i)^2} //$$

Question 2 (C)

In a depth 1 decision tree there are only two leaves in the tree, so we see mainly two classifications.

Additionally, we can see that the fixed step size approach is produces a worse fit compared to the adaptive step size.



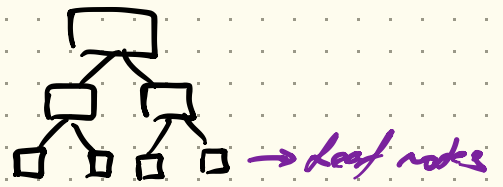

```

1  from sklearn.tree import DecisionTreeRegressor
2  import numpy as np
3  import matplotlib.pyplot as plt
4
5  # true function
6  def f(x):
7      t1 = np.sqrt(x * (1-x))
8      t2 = (2.1 * np.pi) / (x + 0.05)
9      t3 = np.sin(t2)
10     return t1*t3
11
12 def f_sampler(f, n=100, sigma=0.05):
13     # sample points from function f with Gaussian noise (0,sigma**2)
14     xvals = np.random.uniform(low=0, high=1, size=n)
15     yvals = f(xvals) + sigma * np.random.normal(0,1,size=n)
16
17     return xvals, yvals
18
19 def perform_gradient_combination(x, y, T, adaptive):
20
21     # Range of x-values to perform fitted model on
22     xx = np.linspace(0, 1, len(y))
23     xx = xx.reshape(-1, 1)
24
25     # Initial
26     f_prev = np.zeros(len(y))
27
28     # Number of desired base learners
29     for t in range(0, T):
30
31         # (GC1) Residual 'r'
32         # (GC2) Pseudo Data Set 'D'
33         D = np.zeros(len(y))
34         for i in range(0, len(y)):
35             r = y[i] - f_prev[i]
36             D[i] = r
37
38         # (GC3) Fit a model to this residual using our base class F
39         # Prepare machine learning base class F (decision tree that splits populations to two sub-groups for best fit)
40         h = DecisionTreeRegressor(max_depth=2)
41         # Perform the base class on residuals to get a model
42         h = h.fit(x, D)
43         # Perform the tested model on the x values
44         h = h.predict(xx)
45
46         alpha = 0
47         for i in range(0, len(y)):
48             alpha += h[i]
49         alpha /= len(h)
50
51         if (adaptive):
52             f_curr = f_prev + alpha * h
53         else:
54             f_curr = f_prev + 0.1 * h
55         f_prev = f_curr
56
57     return f_curr
58
59
60
61
62 np.random.seed(123)
63 X, y = f_sampler(f, 100, sigma=0.2)
64 X = X.reshape(-1,1)
65
66 # ADAPTIVE
67 fig1, axes1 = plt.subplots(nrows=5, ncols=2, layout='constrained')
68 axes1 = axes1.flatten() # to enumerate through
69
70 for i, ax in enumerate(axes1):
71     adaptive = True
72     base_learner = (i + 2) * 5
73     fitted_model = perform_gradient_combination(X, y, base_learner, adaptive)
74     xx = np.linspace(0,1,1000)
75     ax.plot(xx, f(xx), alpha=0.5, color='red', label='truth')
76     ax.scatter(X,y, marker='x', color='blue', label='observed')
77     ax.plot(np.linspace(0, 1, len(fitted_model)), fitted_model, color='green', label='dt')
78     ax.set_title(f"Plot F: {i*5} (Fixed)")
79
80
81
82 fig2, axes2 = plt.subplots(nrows=5, ncols=2, layout='constrained')
83 axes2 = axes2.flatten() # to enumerate through
84
85 for i, ax in enumerate(axes2):
86     adaptive = False
87     base_learner = (i + 2) * 5
88     fitted_model = perform_gradient_combination(X, y, base_learner, adaptive)
89     xx = np.linspace(0,1,1000)
90     ax.plot(xx, f(xx), alpha=0.5, color='red', label='truth')
91     ax.scatter(X,y, marker='x', color='blue', label='observed')
92     ax.plot(np.linspace(0, 1, len(fitted_model)), fitted_model, color='green', label='dt')
93     ax.set_title(f"Plot F: {i*5} (Adaptive)")
94
95 plt.tight_layout()
96 plt.show()

```

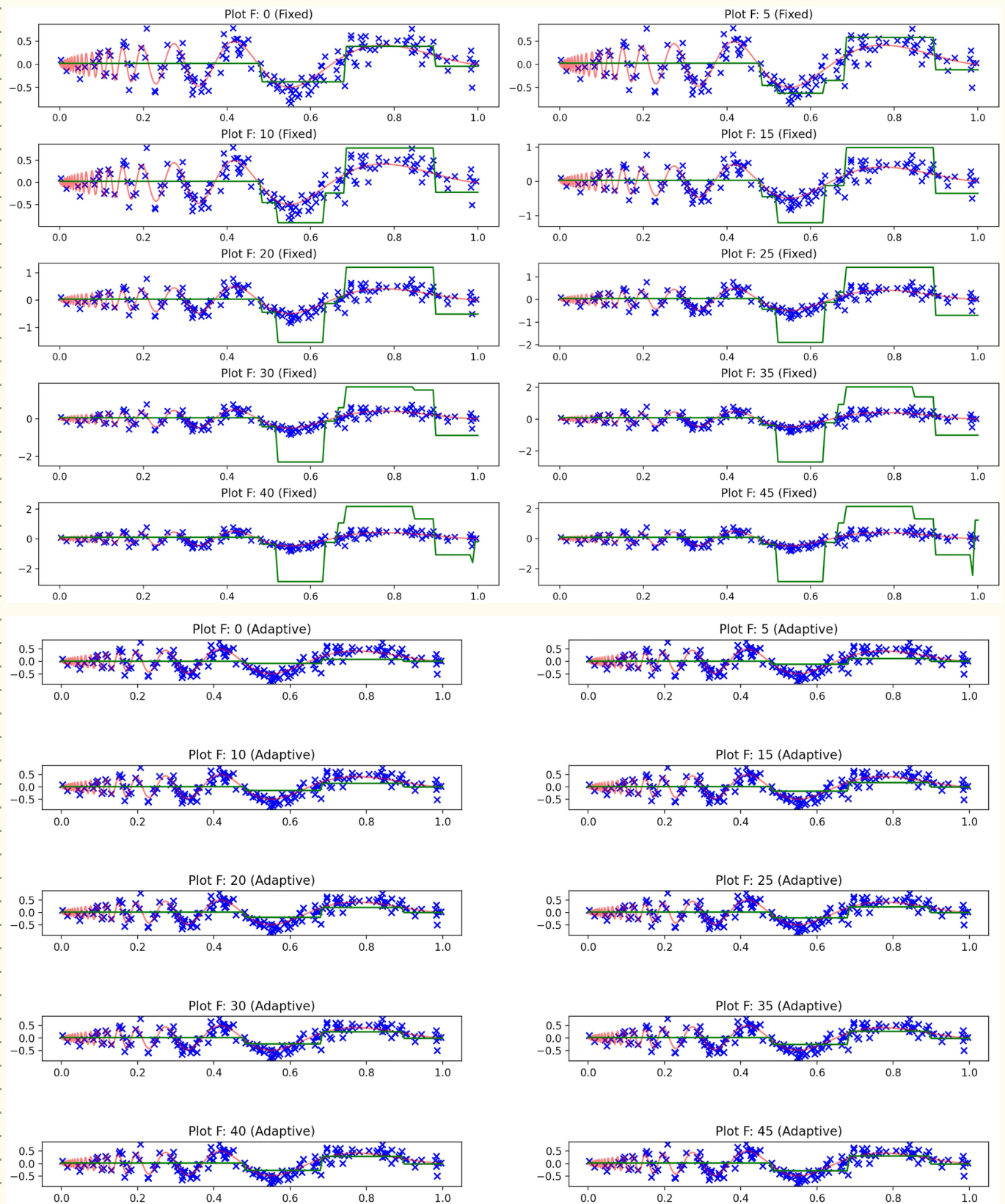
Question 2 (d)

In a tree of depth 2, we can see 4 leaf nodes



From observing the fixed and adaptive method, we can see that, for a large number of base learners, the adaptive model is able to fit the observed data set better.

The fixed step size cannot adjust to changes in residuals, whilst the adaptive approach reduces its step size as the residual decreases.



```

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32         # (GC2) Pseudo Data Set 'D'
33         D = np.zeros(len(y))
34         for i in range(0, len(y)):
35             r = y[i] - f_prev[i]
36             D[i] = r
37
38         # (GC3) Fit a model to this residual using our base class F
39         # Prepare machine learning base class F (decision tree that splits populations to two sub-groups for best fit)
40         h = DecisionTreeRegressor(max_depth=2)
41         # Perform the base class on residuals to get a model
42         h = h.fit(x, D)
43         # Perform the tested model on the x values
44         h = h.predict(xx)
45
46         alpha = 0
47         for i in range(0, len(y)):
48             alpha += h[i]
49         alpha /= len(h)
50
51         if (adaptive):
52             f_curr = f_prev + alpha * h
53         else:
54             f_curr = f_prev + 0.1 * h
55         f_prev = f_curr
56
57     return f_curr
58
59
60
61 np.random.seed(123)
62 X, y = f_sampler(f, 160, sigma=0.2)
63 X = X.reshape(-1,1)
64
65 # ADAPTIVE
66 fig1, axes1 = plt.subplots(nrows=5, ncols=2, layout='constrained')
67 axes1 = axes1.flatten() # to enumerate through
68
69 for i, ax in enumerate(axes1):
70     adaptive = True
71     base_learner = (i + 2) * 5
72     fitted_model = perform_gradient_combination(X, y, base_learner, adaptive)
73     xx = np.linspace(0,1,1000)
74     ax.plot(xx, f(xx), alpha=0.5, color='red', label='truth')
75     ax.scatter(X,y, marker='x', color='blue', label='observed')
76     ax.plot(np.linspace(0, 1, len(fitted_model)), fitted_model, color='green', label='dt')
77     ax.set_title(f"Plot F: {i*5} (Fixed)")
78
79
80
81 fig2, axes2 = plt.subplots(nrows=5, ncols=2, layout='constrained')
82 axes2 = axes2.flatten() # to enumerate through
83
84 for i, ax in enumerate(axes2):
85     adaptive = False
86     base_learner = (i + 2) * 5
87     fitted_model = perform_gradient_combination(X, y, base_learner, adaptive)
88     xx = np.linspace(0,1,1000)
89     ax.plot(xx, f(xx), alpha=0.5, color='red', label='truth')
90     ax.scatter(X,y, marker='x', color='blue', label='observed')
91     ax.plot(np.linspace(0, 1, len(fitted_model)), fitted_model, color='green', label='dt')
92     ax.set_title(f"Plot F: {i*5} (Adaptive)")
93
94
95 plt.tight_layout()
96 plt.show()

```

Question 2 (e)

$$y \in \{-1, 1\}$$

$$\ell(y, \hat{y}) = \log(1 + e^{-y\hat{y}})$$

$$r_{1,i} = -\frac{d}{df(n_i)} \sum_{j=2}^{\hat{n}} \ell(y_j, f(n_j)) \Big|_{f(n_j) = f_{t-1}(n_j), j=2, \dots, \hat{n}}$$

$$= -\frac{d}{df(n_i)} [\log(1 + e^{-y_2 f(n_2)}) + \dots + \log(1 + e^{-y_i f(n_i)}) + \dots + \log(1 + e^{-y_{\hat{n}} f(n_{\hat{n}})})] \Big|_{f(n_j) = f_{t-1}(n_j)}$$

$$= -\frac{d \log[1 + e^{-y_i f(n_i)}]}{df(n_i)} \Big|_{f(n_i) = f_{t-1}(n_i)}$$

$$\text{let } h(x) = 1 + e^{-y_i f(n_i)}$$

$$g(h(x)) = \log[h(x)]$$

$$h'(x) = -y_i f(n_i) e^{-y_i f(n_i)}$$

$$g'(h(x)) = \frac{1}{h(x)}$$

$$= \frac{y_i f(n_i) e^{-y_i f(n_i)}}{1 + e^{-y_i f(n_i)}} \Big|_{f(n_i) = f_{t-1}(n_i)}$$

$$= \frac{y_i f_{t-1}(n_i) e^{-y_i f_{t-1}(n_i)}}{1 + e^{-y_i f_{t-1}(n_i)}}$$

$$1 + e^{-y_i f_{t-1}(n_i)}$$

$$h_t = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^{\hat{n}} \ell(r_{1,i}, f(n_i))$$

$$= \arg \min_{f \in \mathcal{F}} \sum_{i=1}^{\hat{n}} \log \left[1 + e^{-\frac{y_i f_{t-1}(n_i) e^{-y_i f_{t-1}(n_i)}}{1 + e^{-y_i f_{t-1}(n_i)}} \cdot f(n_i)} \right]$$

Question 2 (f)

$$\alpha_+ = \arg \min_{\alpha} \sum_{i=1}^n \ell(y_i, f_{+1}(x_i) + \alpha h_+(x_i))$$

$$= \arg \min_{\alpha} \sum_{i=1}^n \log [1 + e^{-y_i [f_{+1}(x_i) + \alpha h_+(x_i)]}]$$

$$h(u) = 1 + e^{-y_i [f_{+1}(x_i) + \alpha h_+(x_i)]} \quad g(h(u)) = \log(h(u))$$

$$h'(u) = -y_i [f_{+1}(x_i) + \alpha h_+(x_i)] \quad g'(h(u)) = \frac{1}{h(u)}$$

$$\frac{d\alpha_+}{d\alpha} = \sum_{i=1}^n \frac{-y_i [f_{+1}(x_i) + \alpha h_+(x_i)]}{1 + e^{-y_i [f_{+1}(x_i) + \alpha h_+(x_i)]}} = 0$$

Difficult to form a closed form since $\sum \frac{a_i}{b_i} \neq \frac{\sum a_i}{\sum b_i}$

Question 2 (g)

Since we expect the residual to decrease after every iteration, the step size ' α ' required to fit the model of the observed data is also expected to decrease.

A simple algorithm I would implement is to begin the step size at 1.0 i.e. $\alpha = 1$. Then, divide the step size by 2 after every iteration i.e. $\alpha_t = \frac{\alpha_{t-1}}{2}$.

For this algorithm, the only additional computational cost is a division operation.

APPENDIX

KEY

□ Notes & Theorems

□ Derivations

□ Equations

Collection of derivations that were used in the questions. Separated for better readability.

Problems will reference any of these derivations by stating the number label e.g. "according to [1]".

Question 1: $X_1, X_2, \dots, X_n \stackrel{i.i.d}{\sim} N(\mu, \sigma^2)$

[1] $E[\bar{X}] = E[c \cdot S] = \mu$ given that $c \cdot S$ is an unbiased estimator of μ in the question

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i \quad \rightarrow \quad E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n E[X_i] \quad \rightarrow \quad E[X_1 + X_2] = E[X_1] + E[X_2], \quad E[aX] = aE[X] \\ &= \frac{1}{n} \sum_{i=1}^n \mu \quad \rightarrow \quad \sum_{i=1}^n 1 = n \\ &= \mu //\end{aligned}$$

[2] $E[X_i^2] = \sigma^2 + \mu^2$ $Var[X] = E[X^2] - E[X]^2$

$$\begin{aligned}&= Var[X_i] + E[X_i]^2 \\ &= \sigma^2 + \mu^2 //\end{aligned}$$

[3] $Var[\bar{X}] = \frac{\sigma^2}{n}$

$$\begin{aligned}&= Var\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \quad \rightarrow \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \\ &= \frac{1}{n^2} \sum_{i=1}^n Var[X_i] \quad \rightarrow \quad Var[aX] = a^2 Var[X] \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 \quad \rightarrow \quad Var[X_i] = \sigma^2 \\ &= \frac{\sigma^2}{n} // \quad \rightarrow \quad \sum_{i=1}^n 1 = n\end{aligned}$$

[4] $E[\bar{X}^2] = \frac{\sigma^2}{n} + \mu^2$

$$\begin{aligned}&= Var[\bar{X}] + E[\bar{X}]^2 \\ &= \frac{\sigma^2}{n} + \mu^2 //\end{aligned}$$

$$[5] \quad E[X_i \bar{X}] = \frac{\sigma^2}{n} + \mu^2$$

$$= E\left[X_i, \frac{1}{n} \sum_{j=1}^n X_j\right]$$

$$= \frac{1}{n} E\left[\sum_{j=1}^n X_i X_j\right] \quad \text{only when } i=j, X_i \text{ \& } X_j \text{ are not independent.}$$

$$= \frac{1}{n} E\left[X_i^2 + \sum_{j \neq i} X_i X_j\right] \quad E[X+Y] = E[X] + E[Y]$$

$$= \frac{1}{n} \left(E[X_i^2] + \sum_{j \neq i} E[X_i X_j] \right)$$

$$= \frac{1}{n} \left(\sigma^2 + \mu^2 + \sum_{j \neq i} E[X_i] E[X_j] \right) \quad \begin{array}{l} X_1, X_2, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2) \\ \text{independent and from the same population.} \end{array}$$

$$= \frac{1}{n} \left(\sigma^2 + \mu^2 + \sum_{j \neq i} \mu \cdot \mu \right) \quad \sum_{j \neq i} 1 = n-1 \quad (\text{skip the } i^{\text{th}} \text{ situation})$$

$$= \frac{1}{n} \left[\sigma^2 + \mu^2 + (n-1)\mu^2 \right]$$

$$= \frac{1}{n} (\sigma^2 + \mu^2 + n\mu^2 - \mu^2)$$

$$= \frac{\sigma^2}{n} + \mu^2 //$$

$$[6] \quad E[(X_i - \bar{X})^2] = \frac{n-1}{n} \sigma^2$$

$$= E[X_i^2 - 2X_i \bar{X} + \bar{X}^2]$$

$$= \{E[X_i^2] - 2E[X_i \bar{X}] + E[\bar{X}^2]\} \quad \text{see [2], [5], [4]}$$

$$= \left\{ \sigma^2 + \mu^2 - 2\left(\frac{\sigma^2}{n} + \mu^2\right) + \frac{\sigma^2}{n} + \mu^2 \right\}$$

$$= \left(\sigma^2 + \mu^2 - \frac{2\sigma^2}{n} - 2\mu^2 + \frac{\sigma^2}{n} + \mu^2 \right)$$

$$= \sigma^2 - \frac{\sigma^2}{n}$$

$$= \frac{n-1}{n} \sigma^2 //$$

$$[7] \quad E[C_*^2 S^2] = C_*^2 \cdot \sigma^2$$

$$= C_*^2 E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right]$$

$$= \frac{C_*^2}{n-1} \sum_{i=1}^n E[(X_i - \bar{X})^2]$$

$$= \frac{C_*^2}{n-1} \cdot \sum_{i=1}^n \frac{n-1}{n} \sigma^2 \quad \text{see [6]}$$

$$= C_*^2 \cdot \sigma^2 //$$

$$[8] \text{Var}[C_* S] = C_*^2 \cdot \sigma^2 - \mu^2$$

$$= \mathbb{E}[C_*^2 S^2] - \mathbb{E}[C_* S]^2$$

) see [7] § [1]

$$= C_*^2 \cdot \sigma^2 - \mu^2 //$$